

Many lysosomal proteins are transported to the lysosome by the cation-independent mannose 6-phosphate receptor (CI-MPR). CI-MPR is a transmembrane protein found in the Golgi apparatus and vesicles. It contains 15 globular domains with similar (but not identical) amino acid sequences on the luminal side of the membrane. Each domain contains up to 3 glycosylation sites and 6–12 cysteine residues that aid in folding.

The ability of CI-MPR to recognize multiple different ligands was studied by fluorescent microarray analysis. Microarray analysis was performed by cross-linking candidate ligands such as plasminogen, insulin-like growth factor 2 (IGF-II), and mannose 6-phosphate (M6P), to glass slides. The slides were then incubated with full-length CI-MPR or recombinantly expressed CI-MPR domains, followed by incubation with an anti-CI-MPR antibody. A fluorescently labeled antibody that binds the first antibody was then added, and the relative fluorescence levels of each candidate ligand were compared. Higher fluorescence levels correspond to greater binding.

Experiment 1

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Experiment 2

Domains 3, 5, and 9 bind carbohydrates. Amino acids in these domains were mutated to determine which are required for binding. Based on this experiment, scientists concluded that Arg, Glu, and Tyr are required for carbohydrate binding by domains 3, 5, and 9 of CI-MPR.

Table 1

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6	1	8	Unknown	-
7	2	6	Unknown	-
8	1	8	Unknown	-
9	2	8	M6P	6.6×10^{-8}
10	0	8	Unknown	-
11	1	8	IGF-II	1.3×10^{-6}
12	2	8	Unknown	-
13	0	12	IGF-II†	†
14	1	7	Unknown	-
15	0	6	Unknown	-

* M6P-GlcNAc is M6P linked to N-acetylglucosamine (GlcNAc) through the phosphate group

† Domain 13 does not bind IGF-II on its own; instead it increases the affinity of domain 11 for IGF-II

Which class of amino acids is LEAST likely to be found on the surface of CI-MPR globular domains?

- ☐ A. Positively charged [9%]
- ☐ B. Polar uncharged [9%]

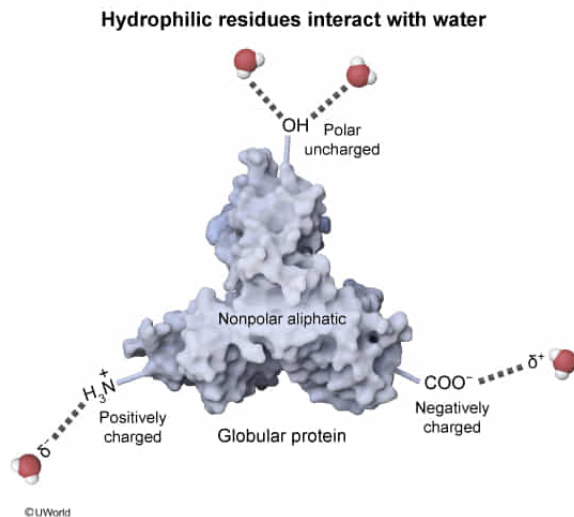
☐ C. Negatively charged [6%]

✓ ☒ D. Nonpolar aliphatic [74%]

Correct

75% answered correctly

Explanation:



Proteins **naturally fold** into their lowest energy conformations as amino acid side chains **optimize their interactions** with each other and with the molecules in their environment. For portions of proteins exposed to aqueous environments, **hydrophilic side chains** interact with polar water molecules through hydrogen bonding and electrostatic interactions.

In contrast, **hydrophobic residues** interact poorly with water. Because they cannot form hydrogen bonds, hydrophobic residues force nearby water molecules to interact with each other to form a highly ordered **solvation layer**. These highly ordered interactions are thermodynamically unfavorable.

According to the passage, the globular domains of CI-MPR are on the **luminal side** of the Golgi apparatus and vesicles. To optimize interactions between these domains and the aqueous environment of the lumen, **polar and charged residues** tend to reside on the **surface**, where they can interact with water. **Nonpolar residues** tend to be found **buried** within the protein or in the hydrophobic environment of a membrane, where they can avoid water. Therefore, nonpolar aliphatic residues are the least likely to be found on the surface of a CI-MPR globular domain.

(Choice A) **Positively charged residues** can interact with the partial negative charge of the oxygen in water molecules. These residues are frequently found on the surface of globular proteins.

(Choice B) **Polar uncharged residues** can hydrogen bond with water and are often found at the surfaces of proteins.

(Choice C) Negatively charged residues can interact with the partial positive charge on the hydrogen atoms in water. They are typically found at the surfaces of proteins.

Educational objective:

Proteins fold into the conformation that optimizes interactions between amino acid residues and their environment. Hydrophilic residues (polar or charged) can interact with water through hydrogen bonding and electrostatic interactions, and they are generally found on the surfaces of globular proteins. Hydrophobic (nonpolar) residues do not interact well with water; they are usually buried inside the protein.

Time spent:0 sec

Subject:
Biochemistry

QID:401880

System:
Amino Acids and Proteins

Many lysosomal proteins are transported to the lysosome by the cation-independent mannose 6-phosphate receptor (CI-MPR). CI-MPR is a transmembrane protein found in the Golgi apparatus and vesicles. It contains 15 globular domains with similar (but not identical) amino acid sequences on the luminal side of the membrane. Each domain contains up to 3 glycosylation sites and 6–12 cysteine residues that aid in folding.

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Which of the following is a covalent bond that could stabilize the tertiary structure of CI-MPR domains?

- ☐ A. Peptide [3%]
- ☒ B. Disulfide [89%]
- ☐ C. Electrostatic [3%]

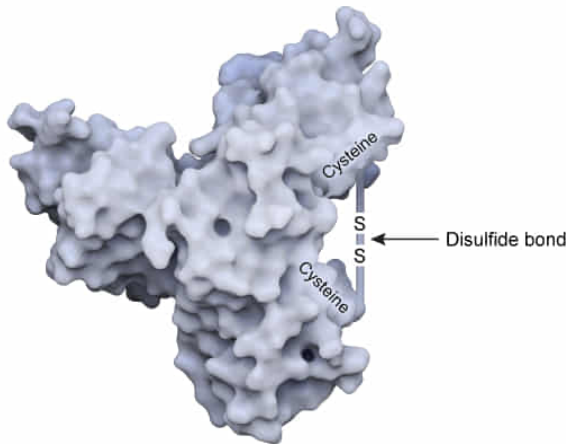
☐ D. Thioester [3%]

Correct

88% answered correctly

Explanation:

Cysteine residues can form disulfide bonds



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A protein's **tertiary structure** is stabilized by interactions between **amino acid side chains**. Most of these **interactions are noncovalent**, including hydrogen bonding and electrostatic attractions. In **oxidizing environments** such as the lumen of vesicles and the extracellular space, covalent bonds can form **between cysteine residues**. Because these bonds form between two sulfur atoms, they are known as **disulfide bonds**.

The passage states that each domain of CI-MPR has between 6 and 12 cysteine residues. Because the globular domains are found in the oxidizing environment of the Golgi apparatus and its vesicles, their tertiary structure is likely stabilized by disulfide bonds between these cysteine residues.

(Choice A) **Peptide bonds** are covalent bonds that form between amino acids to make proteins and peptides. They contribute to a protein's primary structure but not its tertiary structure.

(Choice C) **Electrostatic interactions** are frequently involved in tertiary protein structure, but they do not involve shared electrons and therefore are not covalent bonds.

(Choice D) **Thioester bonds** form between a sulfur atom and a carbonyl carbon. They are important for coenzymes and metabolites such as acetyl-CoA and succinyl-CoA, but they do not participate in tertiary protein structure.

Educational objective:

Tertiary protein structure is stabilized by interactions between amino acid side chains. Most of these interactions are noncovalent, but covalent disulfide bonds may form between cysteine residues in oxidizing environments such as the lumen of a vesicle or the extracellular space.

Time spent:0 sec

Subject:
Biochemistry

QID:401881

System:
Amino Acids and Proteins

Many lysosomal proteins are transported to the lysosome by the cation-independent mannose 6-phosphate receptor (CI-MPR). CI-MPR is a transmembrane protein found in the Golgi apparatus and vesicles. It contains 15 globular domains with similar (but not identical) amino acid sequences on the luminal side of the membrane. Each domain contains up to 3 glycosylation sites and 6–12 cysteine residues that aid in folding.

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Domains 3, 5, and 9 bind carbohydrates. Amino acids in these domains were mutated to determine which are required for binding. Based on this experiment, scientists concluded that Arg, Glu, and Tyr are required for carbohydrate binding by domains 3, 5, and 9 of CI-MPR.

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Based on information in the passage and the data in Table 1, the 15 domains of CI-MPR are most likely similar to each other in which of the following ways?

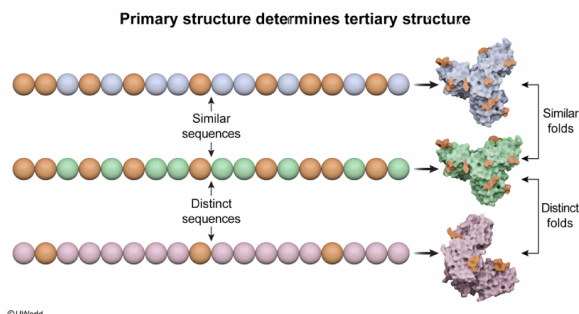
- ☐ A. Post-translational modifications [21%]
- ☐ B. Affinity for their ligands [6%]
- ☒ C. Three-dimensional folded shape [68%]

☐ D. Primary ligand [3%]

Correct

68% answered correctly

Explanation:



The three-dimensional folded form of a protein, including **secondary and tertiary structure**, is entirely **determined by primary structure** (amino acid sequence). Therefore, proteins with **similar amino acid sequences** often **fold similarly**, and those with similar folds most likely have similar amino acid sequences. Proteins with **different sequences** are likely to adopt **distinct folds**. Domains within a protein behave as individual units and typically fold independently from each other.

The passage states that the 15 globular domains of CI-MPR have similar (but not identical) amino acid sequences. Therefore, they most likely all have similar (though not identical) three-dimensional folded shapes.

(Choice A) **Post-translational modifications** are non-amino acid additions to proteins, such as phosphorylation or glycosylation. Table 1 shows that some CI-MPR domains are not glycosylated at all, some have one glycosylation site, some two, and some three. Therefore, the domains are not similar in their post-translational modifications.

(Choice B) The affinity of a protein for its ligand is described by the dissociation constant K_d . Table 1 shows that the different domains have K_d values that differ by several orders of magnitude. Therefore, each domain has a different affinity for its ligand.

(Choice D) Table 1 shows that different domains bind different ligands, and at least one (domain 13) does not bind ligands on its own.

Educational objective:

A protein's secondary and tertiary structures are determined by its amino acid sequence. Proteins with similar amino acid sequences fold similarly, and those with distinct sequences fold differently.

Time spent: 0 sec

Subject:
Biochemistry

QID: 401882

System:
Amino Acids and Proteins

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The data in Table 1 suggest that the domain with the highest affinity for its ligand is:

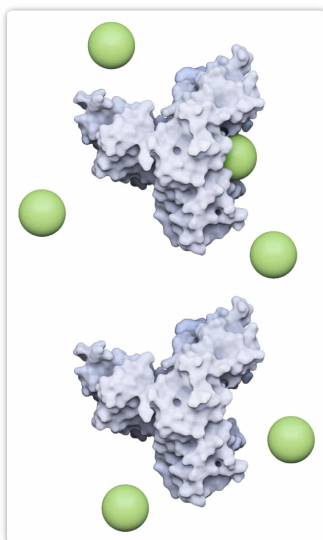
- ☐ A. domain 1. [27%]
- ☐ B. domain 5. [1%]
- ✓ ☒ C. domain 9. [68%]
- ☐ D. domain 11. [2%]

Correct

67% answered correctly

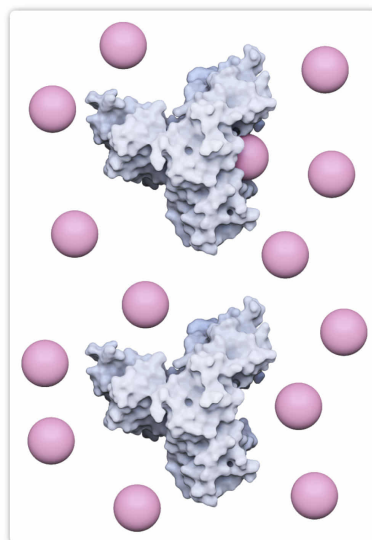
Explanation:

Dissociation constant (K_d) and binding affinity



Low K_d

Few ligands required for binding



High K_d

Many ligands required for binding

The **affinity** of a protein for its ligand is measured by its **dissociation constant K_d** , which quantifies the tendency of the protein-ligand complex to degrade into protein and ligand. Conceptually, K_d represents the concentration of ligand at which half of the proteins in the solution are bound to ligands and half are not. A **small K_d** indicates a low tendency for the complex to break apart and corresponds to a **high affinity** because only a small amount of ligand is required to achieve a high amount of binding.

Table 1 shows that domain 9 binds its ligand (M6P) with a K_d of 6.6×10^{-8} mM, which is approximately half as large as the next smallest K_d of 1.1×10^{-7} mM (domain 3 binding to M6P). Therefore, domain 9 has the highest affinity for its ligand.

(Choices A, B, and D) Domains 1, 5, and 11 all have K_d 's for their ligands that are more than an order of magnitude greater than that of domain 9 for M6P. Therefore, they all have lower affinities for their ligands than domain 9 has for its ligand.

Educational objective:

The affinity of a protein for its ligand is measured by the dissociation constant K_d . A small K_d represents a low tendency of the protein-ligand complex to degrade and a high affinity of the protein for its ligand.

Time spent:0 sec

Subject:
Biochemistry

QID:401883

System:
Amino Acids and Proteins

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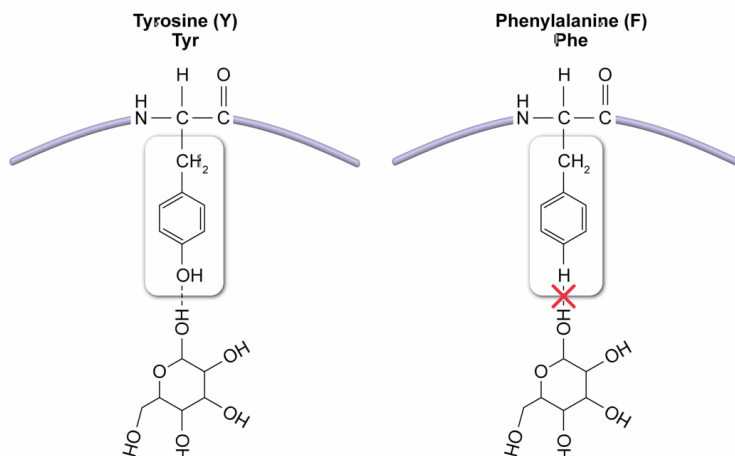
Residues in the binding pockets of domains 3, 5, and 9 interact with carbohydrates through hydrogen bonds. In Experiment 2, tyrosine residues were mutated to remove hydrogen bonding while minimizing changes in the shape of the active site. Tyrosine residues were most likely mutated to:

- ☐ A. Thr [5%]
- ✓ ☒ B. Phe [85%]
- ☐ C. Ala [6%]
- ☐ D. His [2%]

Correct

86% answered correctly

Explanation:



Amino acids can be divided into **different groups** based on the **properties of their side chains**, with many amino acids falling into **more than one group**. For example, tyrosine is predominantly **aromatic** but also has some **polar** characteristics. The polar hydroxyl group on the aromatic ring allows tyrosine to hydrogen bond with other polar molecules such as carbohydrates. Amino acids that are **similar in some ways** and **different in others** can be substituted for each other to preserve some protein properties and change others.

Like tyrosine, phenylalanine (Phe) is aromatic. It is identical to tyrosine except that it has a hydrogen atom where tyrosine has a hydroxyl group. The lack of a hydroxyl group prevents phenylalanine from forming hydrogen bonds, but its otherwise identical structure allows it to occupy similar spaces as tyrosine. It is the best substitute for tyrosine to remove hydrogen bonding without significantly altering the shape of the active site.

(Choice A) **Threonine** (Thr) *can* hydrogen bond and is *not* aromatic. The desired mutation requires an aromatic amino acid that *cannot* hydrogen bond.

(Choice C) **Alanine** (Ala) is not aromatic and is much smaller than tyrosine. Although its substitution for tyrosine would remove hydrogen bonding, it would likely significantly alter the shape of the binding pocket.

(Choice D) **Histidine** (His) is aromatic but polar, and may even be charged in some instances. Its substitution for tyrosine would not remove the ability to hydrogen bond.

Educational objective:

Amino acids can be divided into different groups, with some amino acids falling into more than one group. Substituting one aromatic amino acid for another can preserve aromatic properties. Substituting a polar amino acid for a nonpolar amino acid can

abolish interactions between proteins and ligands.

Time spent:0 sec

Subject:

Biochemistry

QID:401884

System:

Amino Acids and Proteins